

MATH 123 LECTURE NOTES

WEEK ④

FUNCTIONS AND THEIR GRAPHS

→ A function f on a set D into a set S is a rule that assigns a unique element $f(x)$ in S to each element x in D .

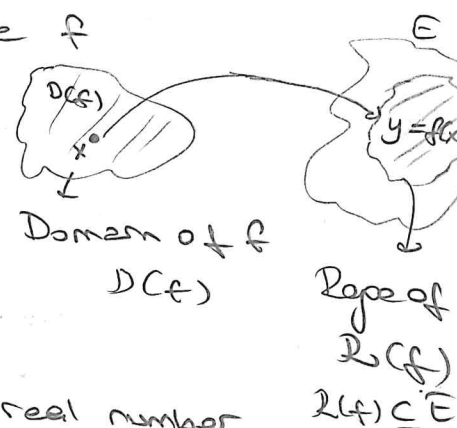
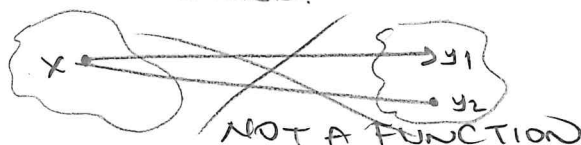
$$f: D(f) \longrightarrow R(f)$$
$$x \longrightarrow y = f(x)$$

$D(f)$: Domain f

$R(f)$: Range f

$$y = f(x)$$

→ Independent variable
→ Dependent variable.



→ Notation The function that converts any real number into its square can be denoted:

• by $y = x^2$

• by $f(x) = x^2$

• by $x \mapsto x^2$

The Domain Convention

→ Let f be a function. Unless specified otherwise, the domain of f is meant to be the set of all real numbers for which $f(x)$ is a real number.

Ex $f(x) = \sqrt{x}$ $D(f) = x \geq 0$

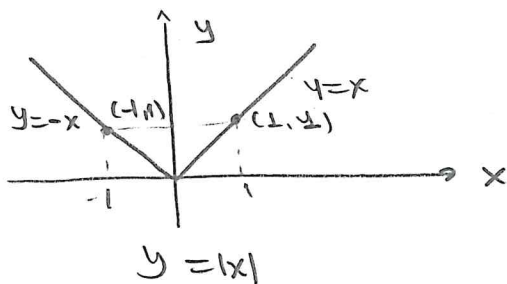
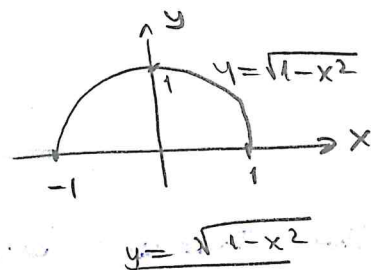
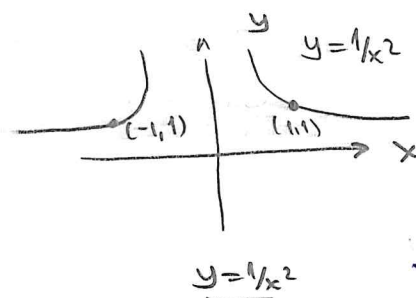
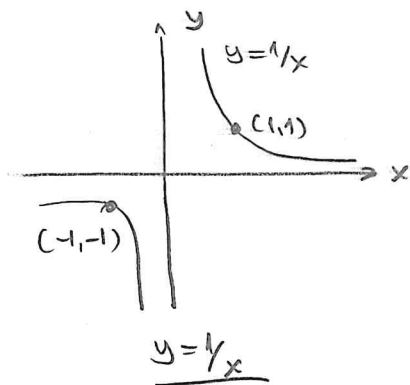
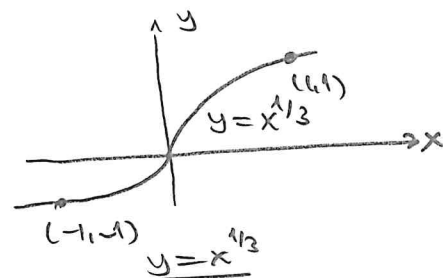
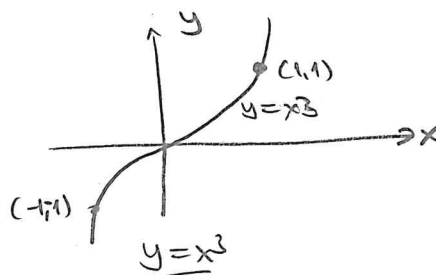
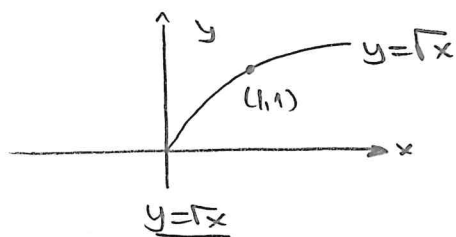
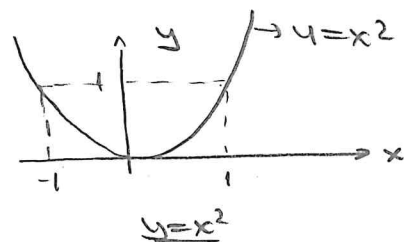
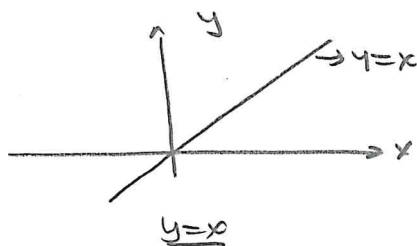
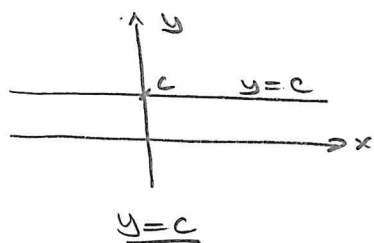
$$f(x) = \frac{x}{x^2 - 4} \quad D(f) = (-\infty, -2) \cup (-2, 2) \cup (2, \infty) = \mathbb{R} \setminus \{-2, 2\}$$

$$f(t) = \sqrt{1 - t^2} \quad 1 - t^2 \geq 0 \quad t^2 \leq 1 \quad -1 \leq t \leq 1 \quad D(f) = [-1, 1]$$

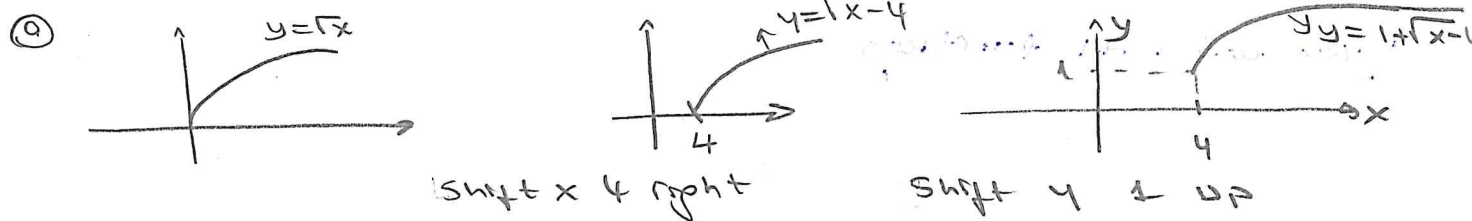
Graphs of Functions

→ The graph of a function f is the set of points (x, y) in the Cartesian plane such that x lies in $D(f)$ and $y = f(x)$

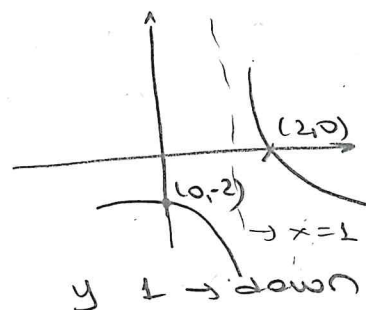
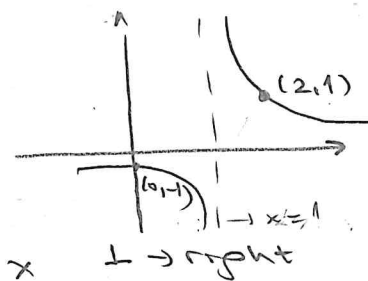
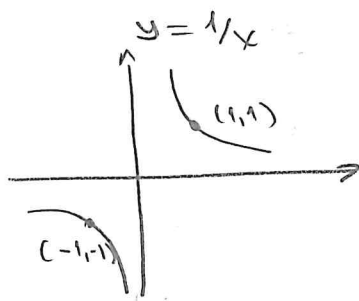
Fundamental Graphs



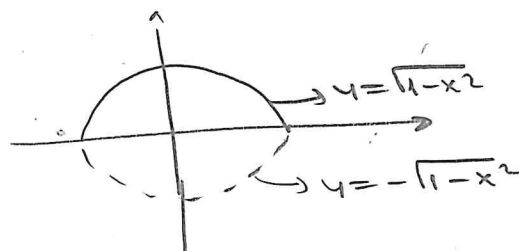
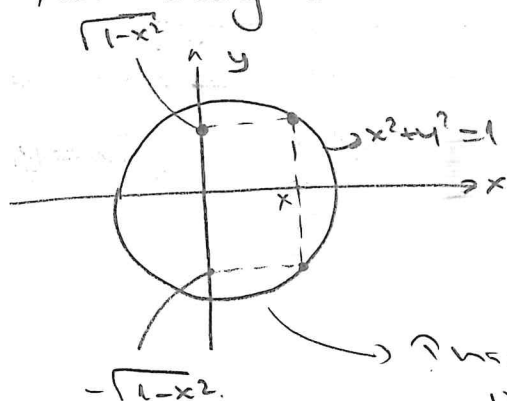
Ex Sketch the graph. (a) $y = 1 + \sqrt{x-4}$ (b) $y = \frac{2-x}{x-1}$



(b) $\frac{2-x}{x-1} = \frac{-x+1+1}{x-1} = -\frac{(x-1)}{x-1} + \frac{1}{x-1} = -1 + \frac{1}{x-1}$



Ex Not every curve is the graph of a function e.g. $x^2 + y^2 = 1$.



$\Rightarrow$ is not a function graph. $\therefore$ is union of the graphs $y = \sqrt{1-x^2}$ and $y = -\sqrt{1-x^2}$.

Even and Odd Functions; Symmetry and Reflections

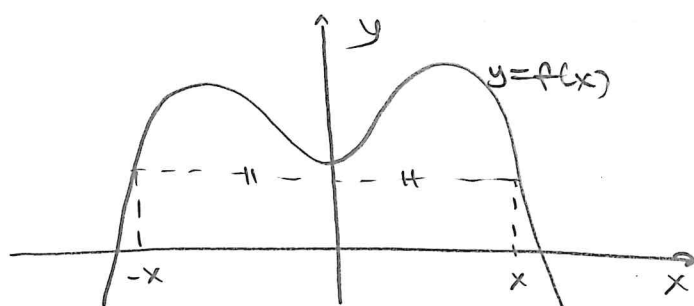
Even and odd functions

→ Suppose that $-x$ belongs to the domain of f whenever x does. We say that f is an even function if

$$f(-x) = f(x) \quad \forall x \in D(f) \quad \text{even function}$$

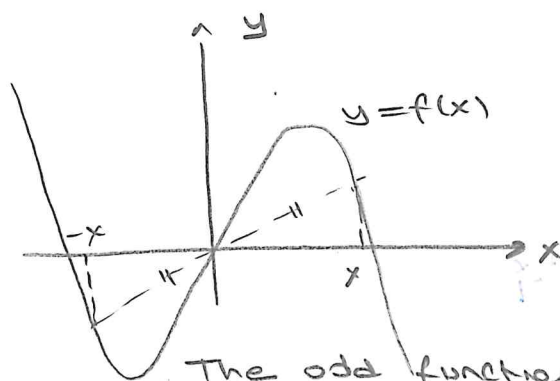
→ We say that f is an odd function if

$$f(-x) = -f(x) \quad \forall x \in D(f) \quad \text{odd function}$$



The even function

Symmetric about y-axis



The odd function.

Symmetric about origin

Reflections in straight lines

Reflections in special lines

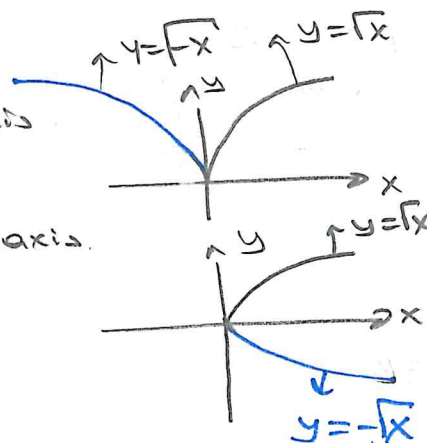
① Substitute $x \rightarrow -x$, Reflect y axis
 $y = \sqrt{x} \rightarrow y = \sqrt{-x}$

② Substitute $y \rightarrow -y$, Reflect x axis.
 $y = \sqrt{x} \rightarrow y = -\sqrt{x}$

③ Substitute $x \rightarrow a-x$, Reflect $x=a/2$

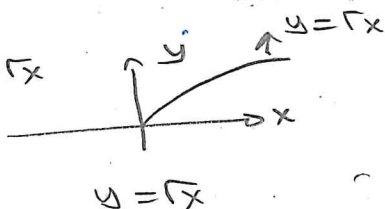
④ Substitute $y \rightarrow b-y$, Reflect $y=b/2$

⑤ Interchange x and y , Reflect $y=x$

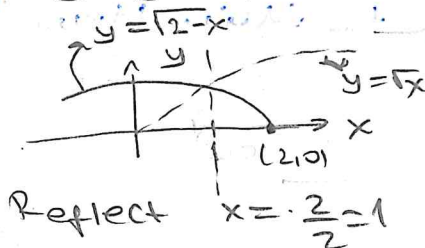


Ex 7 Sketch the graph

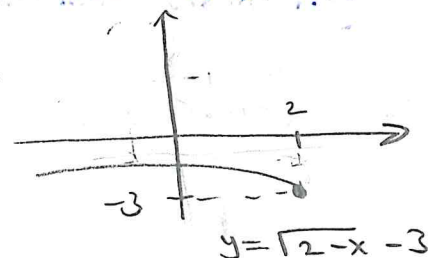
① $y = \sqrt{x}$



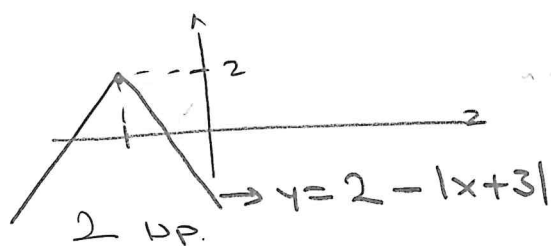
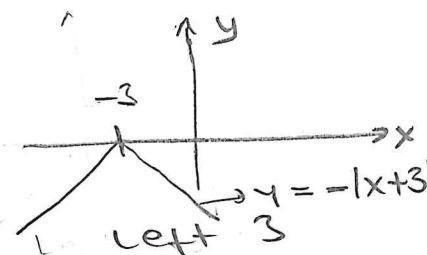
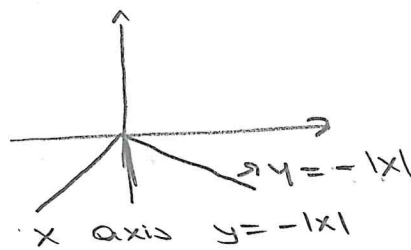
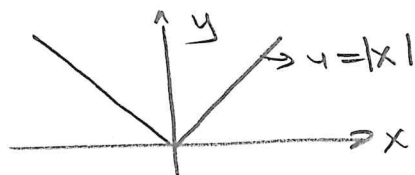
② $y = \sqrt{2-x} - 3$



③ $y = 2 - |x+3|$



④ $y = 2 - |x+3|$



Combining Functions to Make New Functions

Sums, Differences, Products, Quotients and Multiples

→ If f and g are functions, then for every x that belongs to the domains of both f and g we define functions

$$(f+g)(x) = f(x) + g(x)$$

$$(f-g)(x) = f(x) - g(x)$$

$$(fg)(x) = f(x) \cdot g(x)$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad \text{where } g(x) \neq 0$$

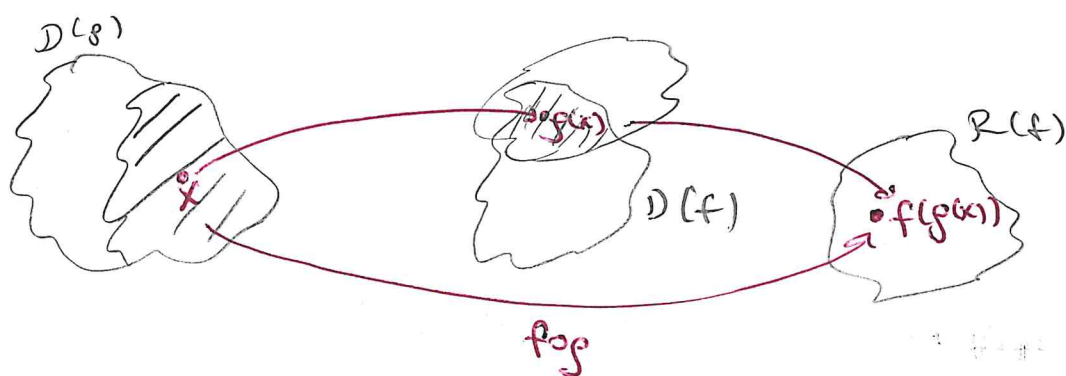
Combining Functions to Make New Functions

Composite Functions

→ If f and g are two functions, the composite function $f \circ g$ is defined by

$$\boxed{f \circ g(x) = f(g(x))}$$

→ The domain of $f \circ g$ consists of those numbers x in the domain of g for which $g(x)$ is in the domain of f . In particular, if the range of g is contained in the domain of f , then the domain of $f \circ g$ is just the domain of g .



Ex Given $f(x) = \sqrt{x}$ $g(x) = x+1$ calculate four composite functions $f \circ g(x)$, $g \circ f(x)$, $f \circ f(x)$, $g \circ g(x)$ and specify the domain of each

$$f(x) = \sqrt{x} \quad D(f) = [0, \infty) \quad g(x) = x+1 \quad D(g) = \mathbb{R}$$

$$f \circ g(x) = f(g(x)) = f(x+1) = \sqrt{x+1} \quad D(f \circ g) = [-1, \infty)$$

$$g \circ f(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{x} + 1 \quad D(g \circ f) = [0, \infty)$$

$$f \circ f(x) = f(f(x)) = f(\sqrt{x}) = \sqrt{\sqrt{x}} = x^{1/4} \quad D(f \circ f) = [0, \infty)$$

$$g \circ g(x) = g(g(x)) = g(x+1) = x+2 \quad D(g \circ g) = \mathbb{R}$$

Ex For $h(x) = \frac{1-x}{1+x}$ $h \circ h(x) = x$. Although the domain of resulting function is $\mathbb{R}$. This is wrong in this case!!! To belong the domain of $h \circ h$, x must satisfy two conditions

- ① x must belong to the domain of h
- ② $h(x)$ must belong to the domain of h

Therefore it must be $\mathbb{R} \setminus \{-1\}$

Piecewise Defined Functions

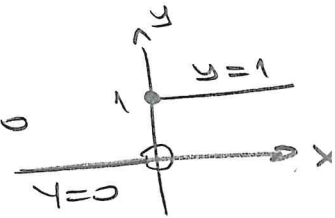
→ Functions defined by different formulas on different parts of the domain.

One simplest example is the absolute value function

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

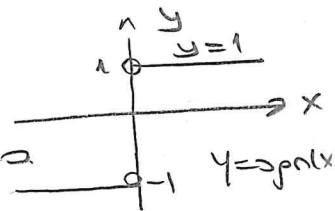
Ex The Heaviside Function

$$H(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$$

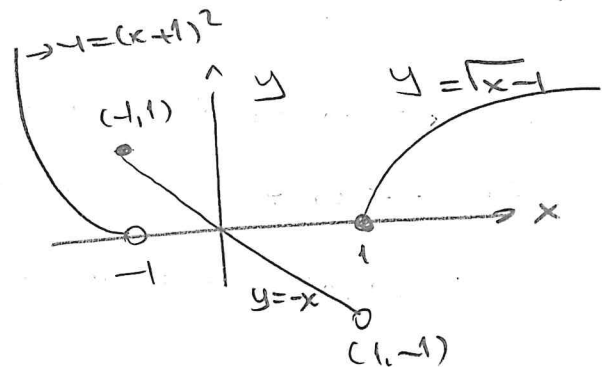


The signum Function

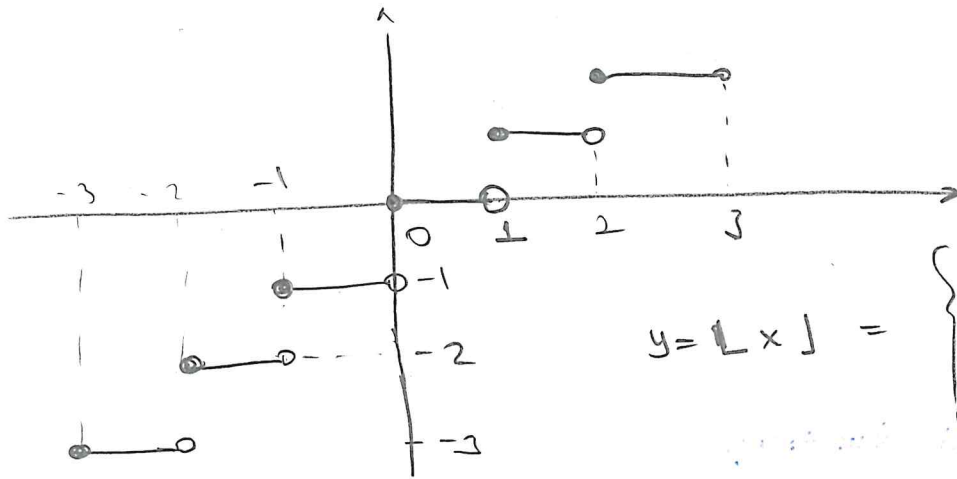
$$\text{sgn}(x) = \frac{x}{|x|} = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \\ \text{undefined} & x = 0 \end{cases}$$



$$f(x) = \begin{cases} (x+1)^2 & \text{if } x < -1 \\ -x & \text{if } -1 \leq x < 1 \\ \sqrt{x-1} & \text{if } x \geq 1 \end{cases}$$

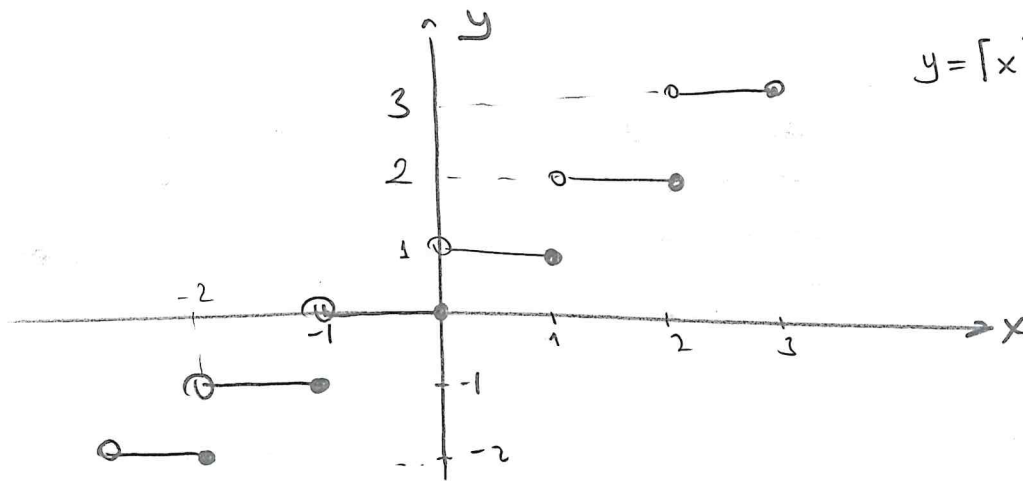


The greatest integer function (Integer floor function) $y = \lfloor x \rfloor$



$$y = \lfloor x \rfloor = \begin{cases} -3 & -3 \leq x < -2 \\ -2 & -2 \leq x < -1 \\ -1 & -1 \leq x < 0 \\ 0 & 0 \leq x < 1 \\ 1 & 1 \leq x < 2 \end{cases}$$

The least integer function (Integer ceiling function) $y = \lceil x \rceil$



$$y = \lceil x \rceil = \begin{cases} -2 & -3 < x \leq -2 \\ -1 & -2 < x \leq -1 \\ 0 & -1 < x \leq 0 \\ 1 & 0 < x \leq 1 \\ 2 & 1 < x \leq 2 \end{cases}$$

Polynomials and Rational Functions

→ A polynomial is a function P whose value at x is

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

$a_n, a_{n-1}, \dots, a_2, a_1, a_0 \rightarrow$ coefficients of the polynomial

$n \rightarrow$ degree of the polynomial

→ A rational function is a quotient $\frac{P(x)}{Q(x)}$

where $P(x)$ and $Q(x)$ are polynomials and $Q(x) \neq 0$

→ If polynomials $P(x)$ and $Q(x)$ are polynomials of degrees m and n , respectively, then we may write

$$\frac{P(x)}{Q(x)} = T(x) + \frac{R(x)}{Q(x)} \rightarrow \text{(Division algorithm)} \quad \text{where } T(x) \text{ and } R(x) \text{ are polynomials and the degree of } R(x) \text{ is less than the degree of } Q(x)$$

Ex 7 Write the division algorithm for $\frac{2x^3 - 3x^2 + 3x + 4}{x^2 + 1}$

$$\begin{array}{r} 2x^3 - 3x^2 + 3x + 4 \quad \overline{) \begin{array}{l} x^2 + 1 \\ 2x - 3 \end{array}} \\ \underline{2x^3 + 2x} \\ -3x^2 + x + 4 \\ \underline{-3x^2 - 3} \\ x + 7 \end{array}$$

$$\frac{2x^3 - 3x^2 + 3x + 4}{x^2 + 1} = 2x - 3 + \frac{x + 7}{x^2 + 1}$$

Roots, zeros and Factors

→ A number r is called a root or zero of a polynomial

$$P(x) \text{ if } P(r) = 0$$

Ex 7 The polynomial $x^3 - x$ has the roots: $0, 1, -1$

The fundamental Theorem of Algebra

Every polynomial of degree at least one has a root
(which might be a complex number)

The Factor Theorem

The number r is a root of the polynomial P of degree not less than 1 if and only if $x-r$ is a factor of $P(x)$.

Corollary Every polynomial of degree $n > 0$ has n roots.

Ex 7 What is the degree of $P(x) = x^3(x^2+2x+5)^2$?
What are the roots of P and, what is multiplicity of each root?

→ If P is expanded, the highest power of x in the expansion is $x^3 \cdot (x^2)^2 = x^7$. Degree 7.

→ $P(x) = x^3(x^2+2x+5)^2$
↓
 $x=0$ root with multiplicity 3

Roots of $x^2+2x+5=0$ $x_{1,2} = \frac{-2 \pm \sqrt{4-4 \cdot 5}}{2} = -1 \pm 2i$

So, we can write $x^2+2x+5 = (x - (-1+2i))(x - (-1-2i))$

$$P(x) = x^3 (x - (-1+2i))^2 (x - (-1-2i))^2$$

$\left\{ \begin{array}{ll} 0, 0, 0 & 0 \text{ with multiplicity } 3 \\ -1-2i, -1-2i & -1-2i \text{ with mult. } 2 \\ -1+2i, -1+2i & -1+2i \text{ with mult. } 2 \end{array} \right.$

Important Factorings

$$\bullet \quad x^2 - a^2 = (x-a)(x+a)$$

$$\bullet \quad x^3 - a^3 = (x-a)(x^2 + ax + a^2)$$

• For an integer $n > 0$:

$$x^n - a^n = (x-a)(x^{n-1} + ax^{n-2} + a^2x^{n-3} + a^3x^{n-4} + \dots + a^{n-2}x + a^{n-1})$$

• For an odd integer $n > 0$:

$$x^n + a^n = (x+a)(x^{n-1} - ax^{n-2} + a^2x^{n-3} - \dots - a^{n-2}x + a^{n-1})$$

Ex Find the roots of the following polynomials

Ⓐ $x^3 - x^2 - 4x + 4$

Ⓑ $x^4 + 3x^2 - 4$

Ⓒ $x^5 - x^4 - x^2 + x$

Ⓐ $x^3 - x^2 - 4x + 4 = x^2(x-1) - 4(x-1) = (x-1)(x^2-4)$ Roots $x = 1, -2, 2$
 $= (x-1)(x+2)(x-2)$

Ⓑ $x^4 + 3x^2 - 4 = \underbrace{(x^2+4)}_{x^2=-4} \underbrace{(x^2-1)}_{x^2=1}$ Roots $x = -1, 1, -2i, 2i$
 $x = \pm 2i \quad x = \pm 1$

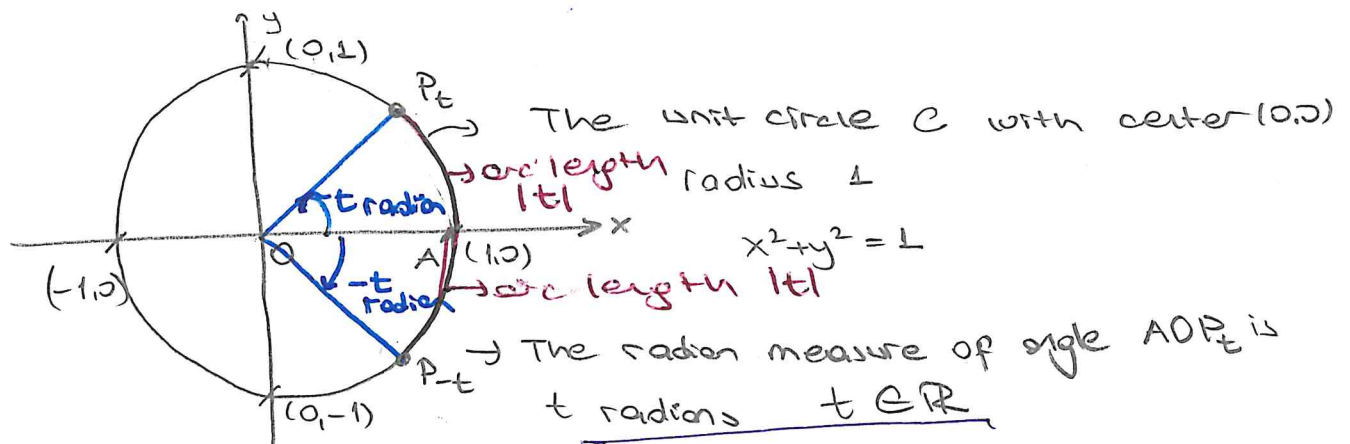
Ⓒ $x^5 - x^4 - x^2 + x = x^4(x-1) - x(x-1) = (x^4-x)(x-1) = x(x^3-1)(x-1)$
 $= x(x-1)(x^2+x+1)(x-1) = x(x-1)^2(x^2+x+1)$

$x = 0, \quad x = 1 \rightarrow \text{multiplicity } 2, \quad x^2+x+1=0 \quad x_{1,2} = \frac{-1 \pm \sqrt{1-4}}{2}$

$$x_{1,2} = \frac{-1 \pm \sqrt{3}i}{2}$$

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THE TRIGONOMETRIC FUNCTIONS



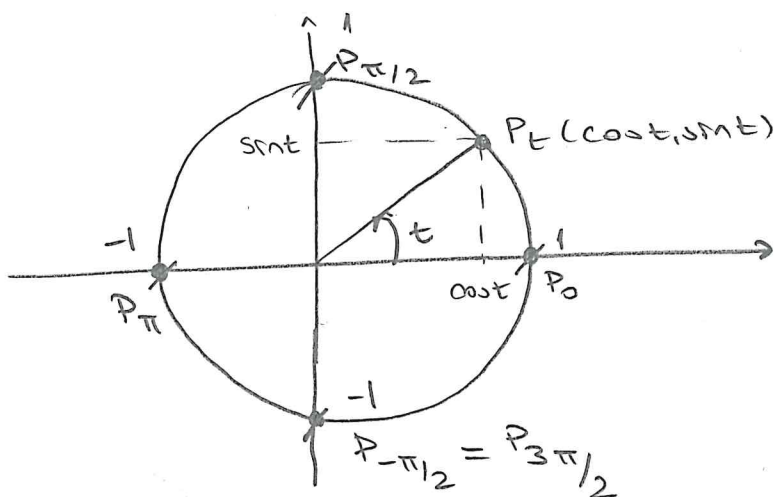
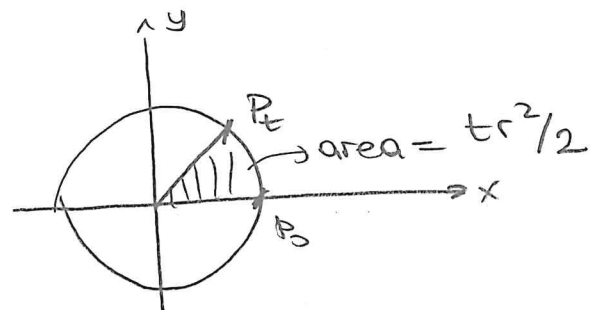
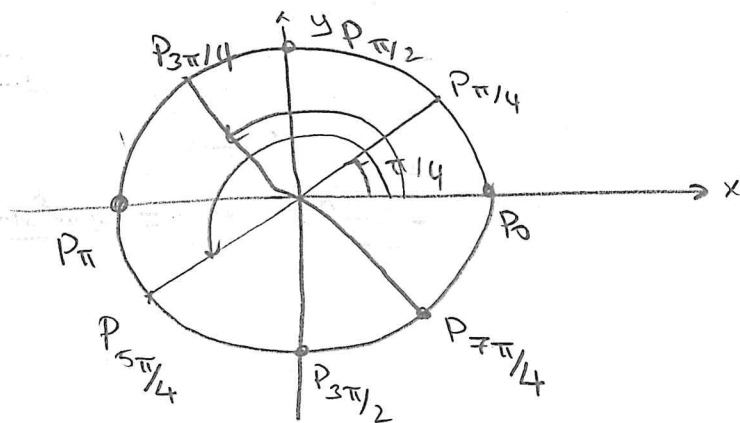
$$AOP_t = t \text{ radians}$$

$$AOP_{-t} = -t \text{ radians}$$

→ we are more used to measuring angles in degrees.

$$\pi \text{ radians} = 180^\circ$$

→ This means that we have



$$\cos 0 = 1 \quad \cos \pi/2 = 0$$

$$\cos \pi = -1 \quad \cos(-\pi/2) = \cos 3\pi/2 = 0$$

$$\sin 0 = 0 \quad \sin \pi/2 = 1$$

$$\sin \pi = 0 \quad \sin(-\pi/2) = \sin 3\pi/2 = -1$$

Properties of Cosine and Sine Function

The range of cosine and sine $-1 \leq \cos t \leq 1$ $-1 \leq \sin t \leq 1$

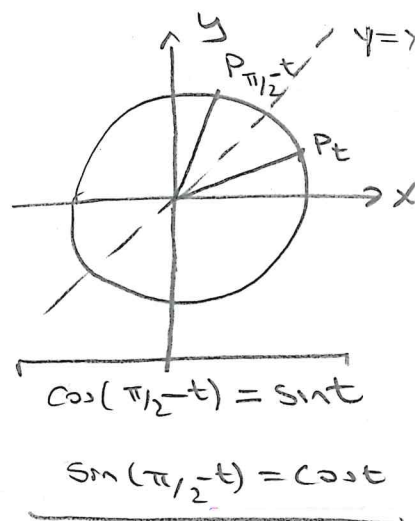
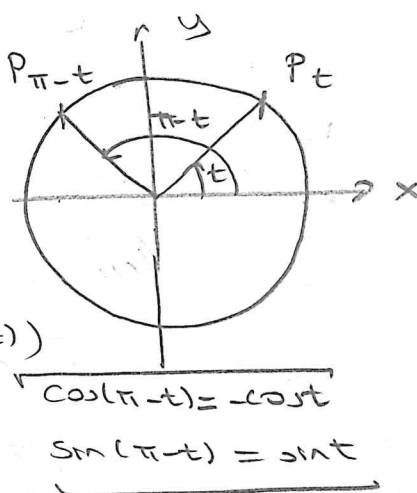
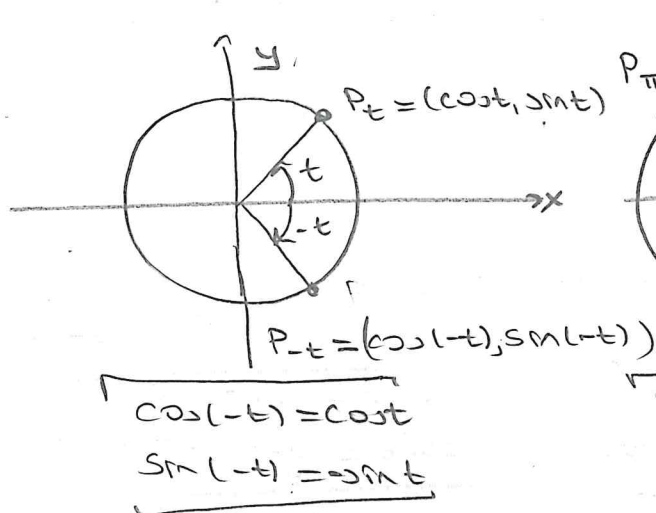
The Pythagorean identity $\cos^2 t + \sin^2 t = 1$

Periodicity $\cos(t + 2\pi) = \cos t$ $\sin(t + 2\pi) = \sin t$

Cosine even, Sine odd $\cos(-t) = \cos t$ $\sin(-t) = -\sin t$

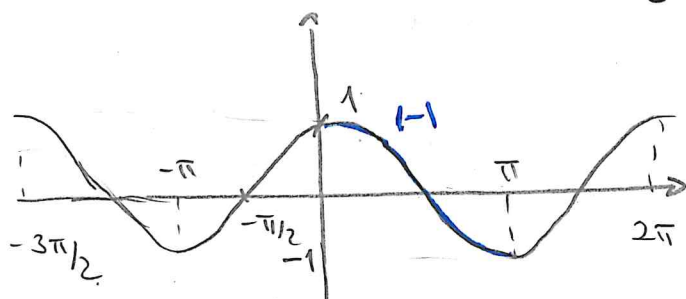
Complementary angle identities $\cos(\frac{\pi}{2} - t) = \sin t$ $\sin(\frac{\pi}{2} - t) = \cos t$

Supplementary angle identities $\cos(\pi - t) = -\cos t$ $\sin(\pi - t) = \sin t$

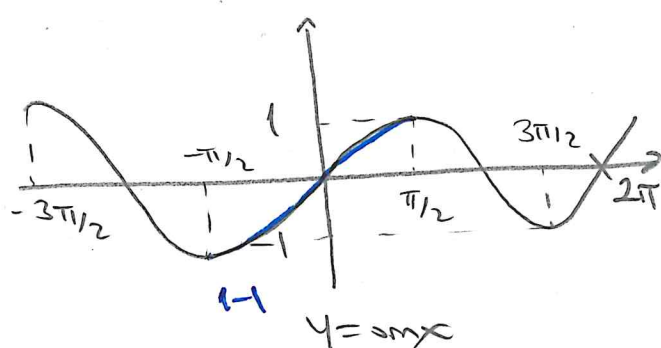


Some Special Angles

Degrees	0°	30°	45°	60°	90°	120°	135°	150°	180°
Radians	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	$3\pi/4$	$5\pi/6$	π
Cosine	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
Sine	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0



$D(\cos x) = \mathbb{R}$



$D(\sin x) = \mathbb{R}$

The Addition Formulas

$$\cos(s+t) = \cos s \cos t - \sin s \sin t$$

$$\sin(s+t) = \sin s \cos t + \cos s \sin t$$

$$\cos(s-t) = \cos s \cos t + \sin s \sin t$$

$$\sin(s-t) = \sin s \cos t - \cos s \sin t$$

Ex Find the value of $\cos(\pi/12) = \cos 15^\circ$.

$$\begin{aligned}\cos(\pi/12) &= \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \cos \pi/3 \cdot \cos \pi/4 + \sin \pi/3 \cdot \sin \pi/4 \\ &= \frac{1}{2} \cdot \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} = \frac{1+\sqrt{3}}{2\sqrt{2}}\end{aligned}$$

$$\sin 2t = 2 \sin t \cos t$$

$$\cos 2t = \cos^2 t - \sin^2 t$$

$$\begin{aligned}\cos 2t &= \cos^2 t - \sin^2 t = \cos^2 t - (1 - \cos^2 t) = 2\cos^2 t - 1 \\ &= 1 - \sin^2 t - \sin^2 t = 1 - 2\sin^2 t\end{aligned}$$

$$\cos^2 t = \frac{\cos 2t + 1}{2}$$

$$\sin^2 t = \frac{1 - \cos 2t}{2}$$

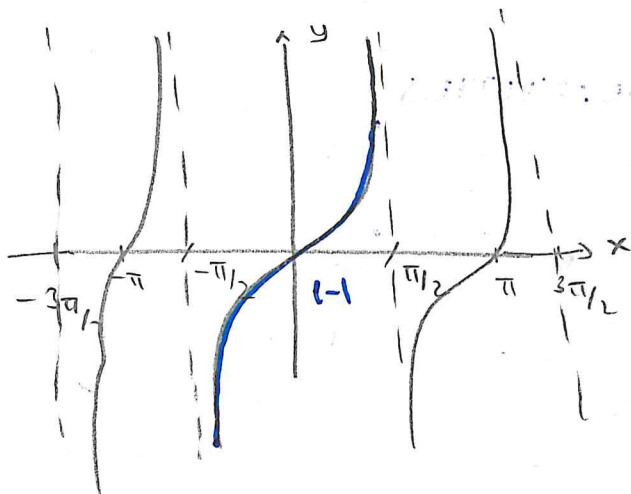
Other Trigonometric Functions

$$\tan t = \frac{\sin t}{\cos t}$$

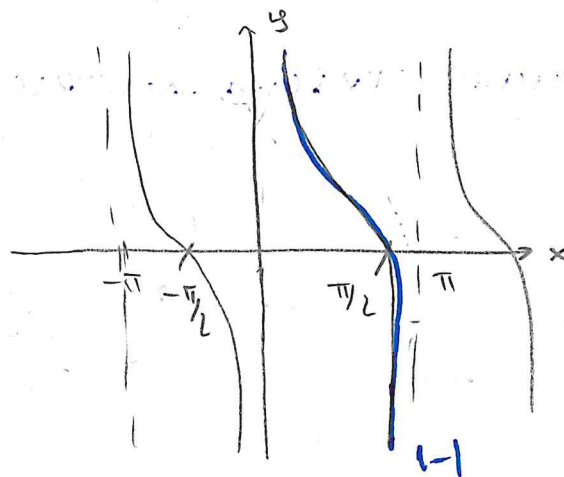
$$\sec t = \frac{1}{\cos t}$$

$$\cot t = \frac{\cos t}{\sin t}$$

$$\operatorname{cosec} t = \frac{1}{\sin t}$$



$$y = \tan x$$



$$y = \cot x$$

$$D(\tan x) = \mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi, k \in \mathbb{Z} \right\}$$

$$R(\tan x) = \mathbb{R}$$

$$\tan(x+\pi) = \tan x$$

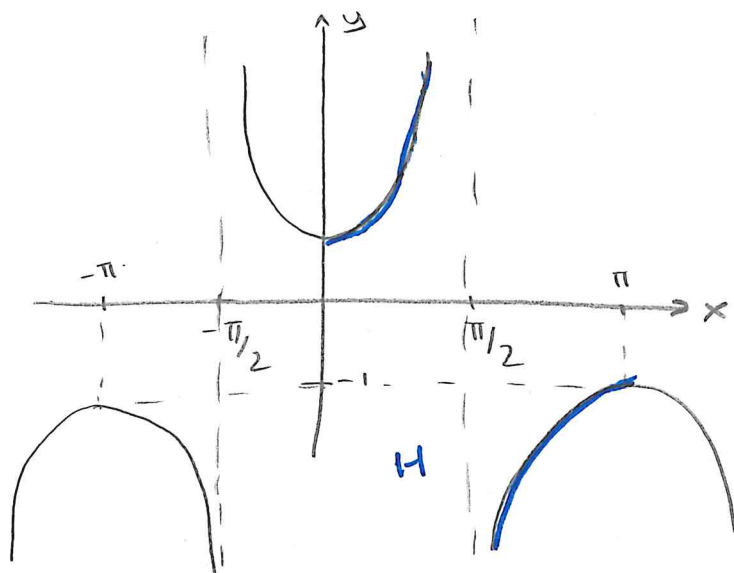
$$\tan(-x) = -\tan x$$

$$D(\cot x) = \mathbb{R} \setminus \{ \pi + k\pi : k \in \mathbb{Z} \}$$

$$R(\cot x) = \mathbb{R}$$

$$\cot(x+\pi) = \cot x$$

$$\cot(-x) = -\cot x$$



$$y = \sec x$$

$$D(\sec x) = \mathbb{R} \setminus \{\pi/2 + k\pi, k \in \mathbb{Z}\}$$

$$R(\sec x) = (-\infty, -1] \cup [1, \infty)$$

$$\sec(-x) = -\sec x$$

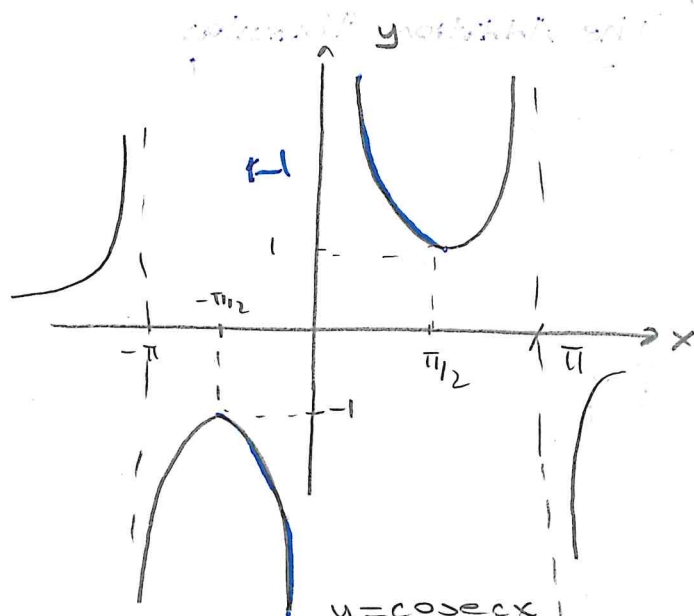
$$\sec(x+2\pi) = \sec x$$

Sine +

A

Tangent + Cotangent

Cosine +



$$y = \csc x$$

$$D(\csc x) = \mathbb{R} \setminus \{k\pi, k \in \mathbb{Z}\}$$

$$R(\csc x) = (-\infty, -1] \cup [1, \infty)$$

Cosec Rule

$$\csc(-x) = -\csc x$$

$$\csc(x+2\pi) = \csc x$$

Ex Find the sine and tangent of the angle θ in $[\pi, 3\pi/2]$ for which $\cos \theta = -1/3$

$$\sin^2 \theta + \underbrace{\cos^2 \theta}_{1/9} = 1$$

$$\sin^2 \theta = 1 - 1/9 = 8/9$$

$$\sin \theta = -\frac{2\sqrt{2}}{3}$$

$$\theta \in [\pi, 3\pi/2] \Rightarrow \sin \theta = -$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{-\frac{2\sqrt{2}}{3}}{-\frac{1}{3}} = 2\sqrt{2}$$

OTHER TRIGONOMETRIC IDENTITIES

$$\tan(x+\pi) = \tan x$$

$$\cot(x+\pi) = \cot x$$

$$1 + \tan^2 x = \sec^2 x$$

$$\tan(s \mp t) = \frac{\tan s \mp \tan t}{1 \pm \tan s \tan t}$$

$$1 + \cot^2 x = \csc^2 x$$

$$1 \pm \tan s \tan t$$

$$\cot(s \mp t) = \frac{\cot s \cot t \pm 1}{\cot t \mp \cot s}$$

$$\tan(x+\pi) = \frac{\sin(x+\pi)}{\cos(x+\pi)} = \frac{\sin x \cos \pi + \cos x \sin \pi}{\cos x \cos \pi - \sin x \sin \pi} = \frac{-\sin x}{-\cos x} = \tan x$$

Thm

Sine Law

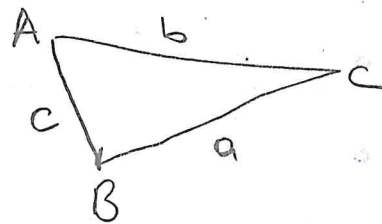
$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Cosine Law

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

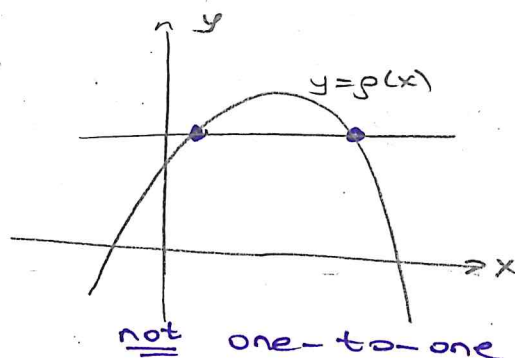
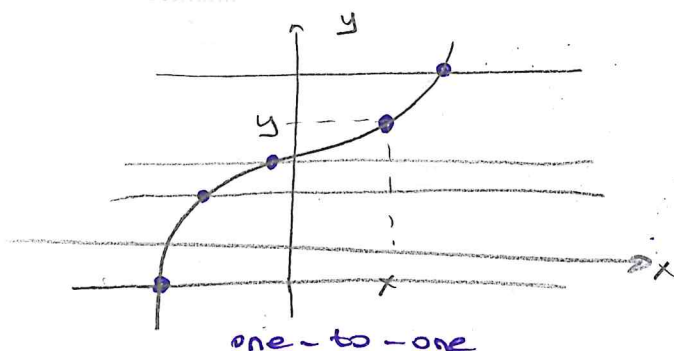
$$c^2 = a^2 + b^2 - 2ab \cos C$$



INVERSE FUNCTIONS

→ A function f is one-to-one if $f(x_1) \neq f(x_2)$ whenever x_1 and x_2 belong to the domain of f and $x_1 \neq x_2$ or, equivalently,

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$



If f is one-to-one, then it has an inverse function f^{-1} . The value of $f^{-1}(x)$ is the unique number y in the domain of f for which $f(y) = x$. Thus,

$$y = f^{-1}(x) \Leftrightarrow x = f(y)$$

Ex Show that $f(x) = 2x - 1$ is one-to-one, and find its inverse $f^{-1}(x)$

$f'(x) = 2 > 0$ it is an increasing function. Therefore, it is one-to-one.

$$y = 2x - 1 \Rightarrow \frac{y+1}{2} = x$$

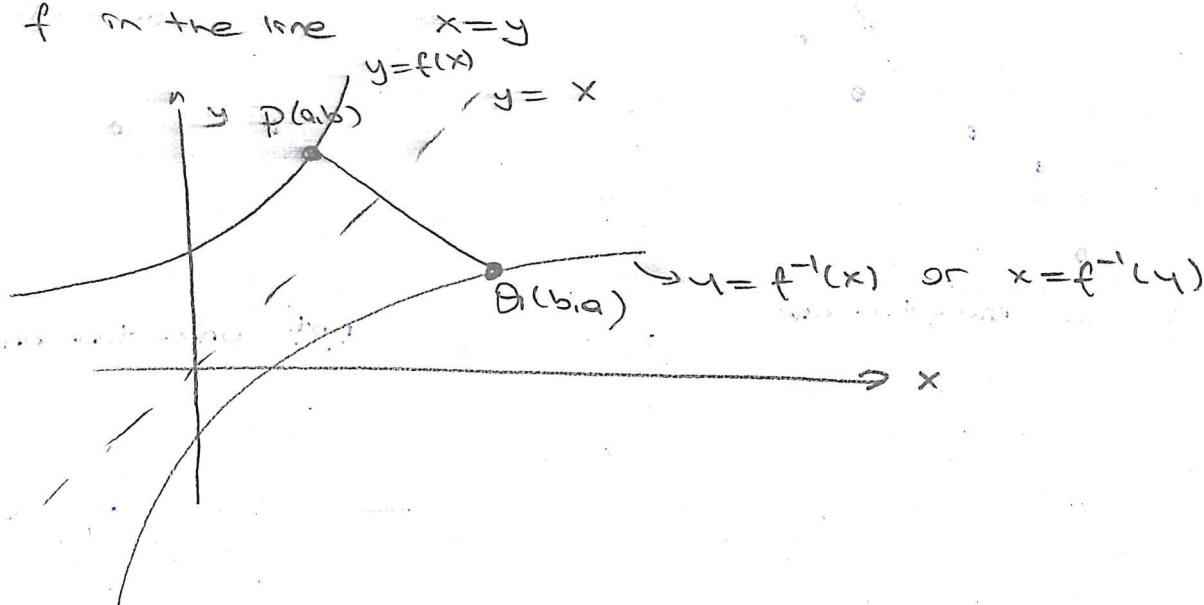
$$y = \frac{x+1}{2}$$

$$f^{-1}(x) = \frac{x+1}{2}$$

Properties of Inverse Functions

- $y = f^{-1}(x) \Leftrightarrow x = f(y)$
- The domain of f^{-1} $D(f^{-1})$ is the range of f $R(f)$. $D(f^{-1}) = R(f)$
- The range of f^{-1} $R(f^{-1})$ is the domain of f $D(f)$. $R(f^{-1}) = D(f)$
- $f^{-1}(f(x)) = x$ for all x in the domain of f
 $\rightarrow f^{-1} \circ f = I_{D(f)}$
- $f(f^{-1}(x)) = x$ for all x in the domain of f^{-1}
 $\rightarrow f \circ f^{-1} = I_{D(f^{-1})}$
- $(f^{-1})^{-1}(x) = f(x)$ for all x in the domain of f .

- The graph of f^{-1} is the reflection of the graph of f in the line $x=y$



Ex 7 Show that $g(x) = \sqrt{2x+1}$ is invertible and find its inverse

If $g(x_1) = g(x_2)$, then $\sqrt{2x_1+1} = \sqrt{2x_2+1}$

$2x_1+1 = 2x_2+1 \Rightarrow x_1 = x_2$ g is one to one ✓

Let $y = g^{-1}(x)$ $g(y) = x \Rightarrow \sqrt{2y+1} = x$

$2y = x^2 - 1$ $y = \frac{x^2-1}{2} = g^{-1}(x)$ for $x \geq 0$
 (17) since $D(g) = [0, \infty)$

Ex 7 Show that $f(x) = \frac{x-1}{x-2}$ is one to one, find its inverse and domain and range of f and f^{-1} .

One to one $f(x_1) = f(x_2) \quad \frac{x_1-1}{x_1-2} = \frac{x_2-1}{x_2-2}$

$$(x_1-1)(x_2-2) = (x_1-2)(x_2-1) \Rightarrow \cancel{x_1 x_2} - 2x_1 - x_2 + \cancel{2} = \cancel{x_1 x_2} - x_1 - 2x_2 + \cancel{2}$$

$$\underline{\underline{x_1 = x_2}}$$

Inverse

$$y = f^{-1}(x) \Rightarrow g(y) = x$$

$$\frac{y-1}{y-2} = x$$

$$x = 1 + \frac{1}{y-2}$$

$$x-1 = \frac{1}{y-2}$$

$$y-2 = \frac{1}{x-1}$$

$$y = \frac{1}{x-1} + 2$$

$$f(x) = 1 + \frac{1}{x-2}$$

$$D(f) = \mathbb{R} \setminus \{2\}$$

$$R(f) = \mathbb{R} \setminus \{1\}$$

Since $\frac{1}{x-2} \neq 0$ we have

$$f^{-1}(x) = 2 + \frac{1}{x-1}$$

$$D(f^{-1}) = \mathbb{R} \setminus \{1\}$$

$$R(f^{-1}) = \mathbb{R} \setminus \{2\}$$

EXPONENTIAL AND LOGARITHMIC FUNCTIONS

→ An exponential function is a function of the form

$$f(x) = a^x$$

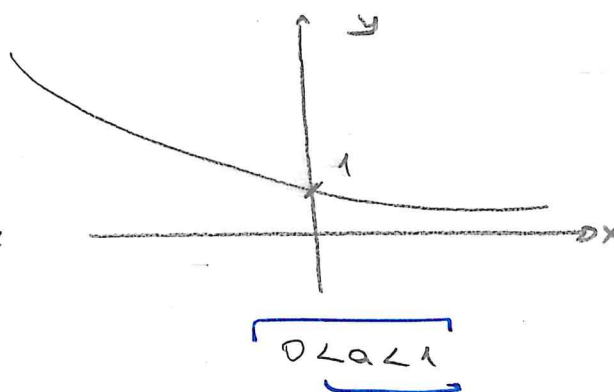
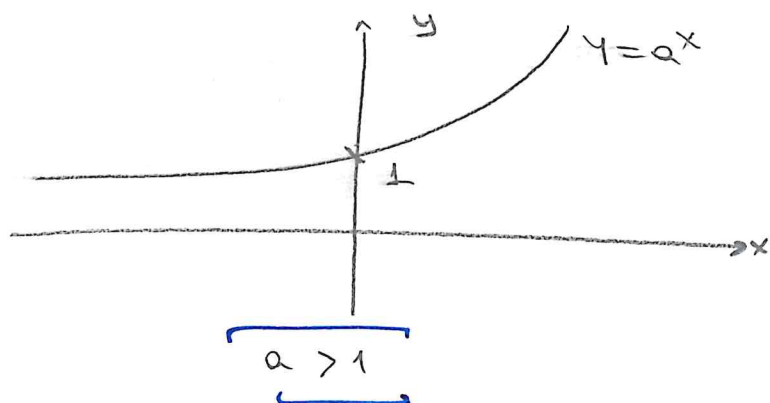
where a is a positive constant which is not 1

→ If x is a positive integer, then $a^x = \underbrace{a \cdot a \cdot a \cdots a}_{x \text{ factors}}$

We assume $a^0 = 1$, and if x is a negative integer then $a^x = \frac{1}{a^{-x}}$

→ If $x = p/q$ is a rational number where p and $q > 0$ are integers, then

$$a^x = a^{p/q} = \sqrt[q]{a^p} = (\sqrt[q]{a})^p$$



Laws of exponents

→ If $a > 0$ and $b > 0$, and x and y are any real numbers, then

(i) $a^0 = 1$

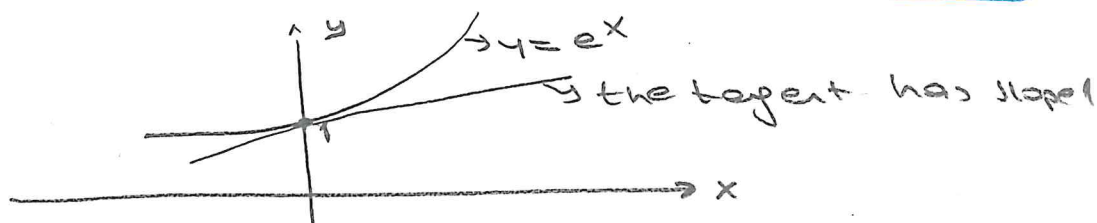
(ii) $a^{x+y} = a^x \cdot a^y$

(iii) $a^{-x} = \frac{1}{a^x}$

(iv) $a^{x-y} = \frac{a^x}{a^y}$

(v) $(a^x)^y = a^{xy}$

(vi) $(ab)^x = a^x b^x$



The Natural Exponential Function

Logarithms

If $a > 0$ and $a \neq 1$, the function $\log_a x$, called the logarithm of x to the base a , is the inverse of the one-to-one function a^x

$$y = \log_a x \iff x = a^y \quad (a > 0, a \neq 1)$$

a^x is a function $\mathbb{R} \mapsto (0, \infty)$

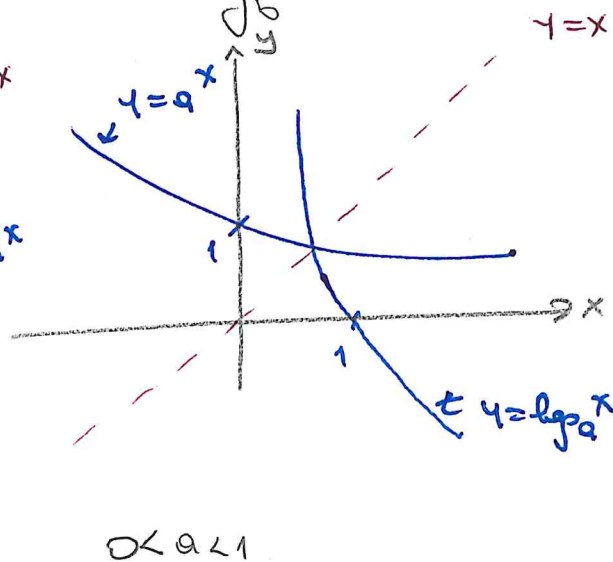
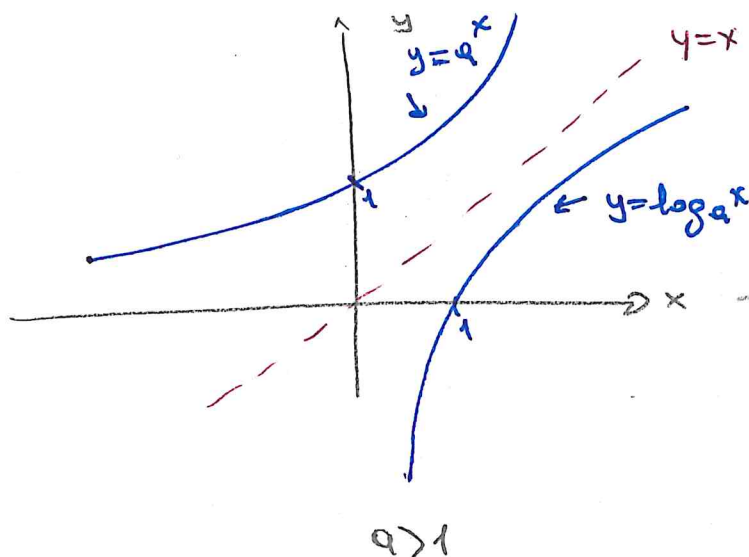
$\log_a x$ is a function $(0, \infty) \mapsto \mathbb{R}$

Properties of Logarithms

- $\log_a(a^x) = x$ for all real x
- $a^{\log_a x} = x$ for all $x > 0$

If $x > 0, y > 0, a > 0, b > 0, a \neq 1$ and $b \neq 1$, then

- $\log_a 1 = 0$
- $\log_a\left(\frac{1}{x}\right) = -\log_a x$
- $\log_a(x^y) = y \log_a x$
- $\log_a(xy) = \log_a x + \log_a y$
- $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$
- $\log_a x = \frac{\log_b x}{\log_b a}$



Ex 7 If $a > 0, x > 0, y > 0$ verify that $\log_a(xy) = \log_a x + \log_a y$

Let $u = \log_a x$ $v = \log_a y$. By the definition of inverse function:

$x = a^u$ $y = a^v$, then $xy = a^u \cdot a^v = a^{u+v}$. This means that

$$\log_a(xy) = u + v = \log_a x + \log_a y$$

Ex 7 Simplify (a) $\log_2 10 + \log_2 12 - \log_2 15$ (b) $\log_{a^2} a^3$ (c) $3^{\log_9 4}$

$$(a) \log_2 10 + \log_2 12 - \log_2 15 = \log_2 \frac{10 \cdot 12}{15} = \log_2 8 = \log_2 2^3 = \underline{3}$$

$$(b) \log_{a^2} a^3 = 3 \log_{a^2} a = \frac{3}{2} \log_{a^2} a^2 = \underline{\frac{3}{2}}$$

$$(c) 3^{\log_9 4} = 3^{(\log_3 4) / (\log_3 9)} = (3^{\log_3 4})^{1/\log_3 9} = (4)^{1/\log_3 3^2} = 4^{1/2} = \underline{2}$$

Ex 7 Solve the equation $3^{x-1} = 2^x$

Take logarithm of both sides

$$(x-1) \log_a 3 = x \log_a 2$$

$$(\log_a 3 - \log_a 2) x = \log_a 3$$

$$x = \frac{\log_a 3}{\log_a 3/2}$$